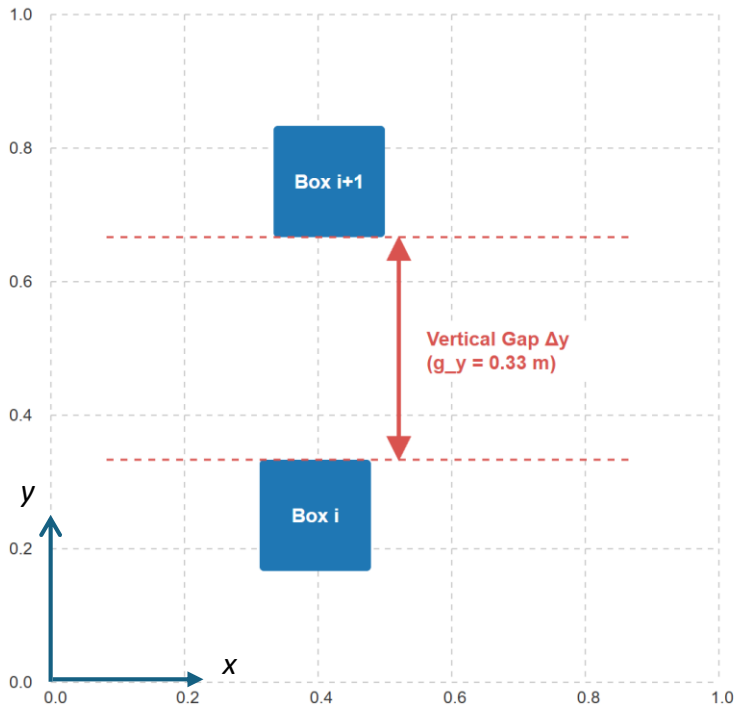
POLITECNICO MILANO 1863 | DEPARTMENT OF MECHANICAL ENGINEERING

PROJECT PRESENTATION:
Machine learning for mechanical systems

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Course: **Machine learning for mechanical systems**– A.Y. 2025/26

Project 1: Sorting packages (Environment)

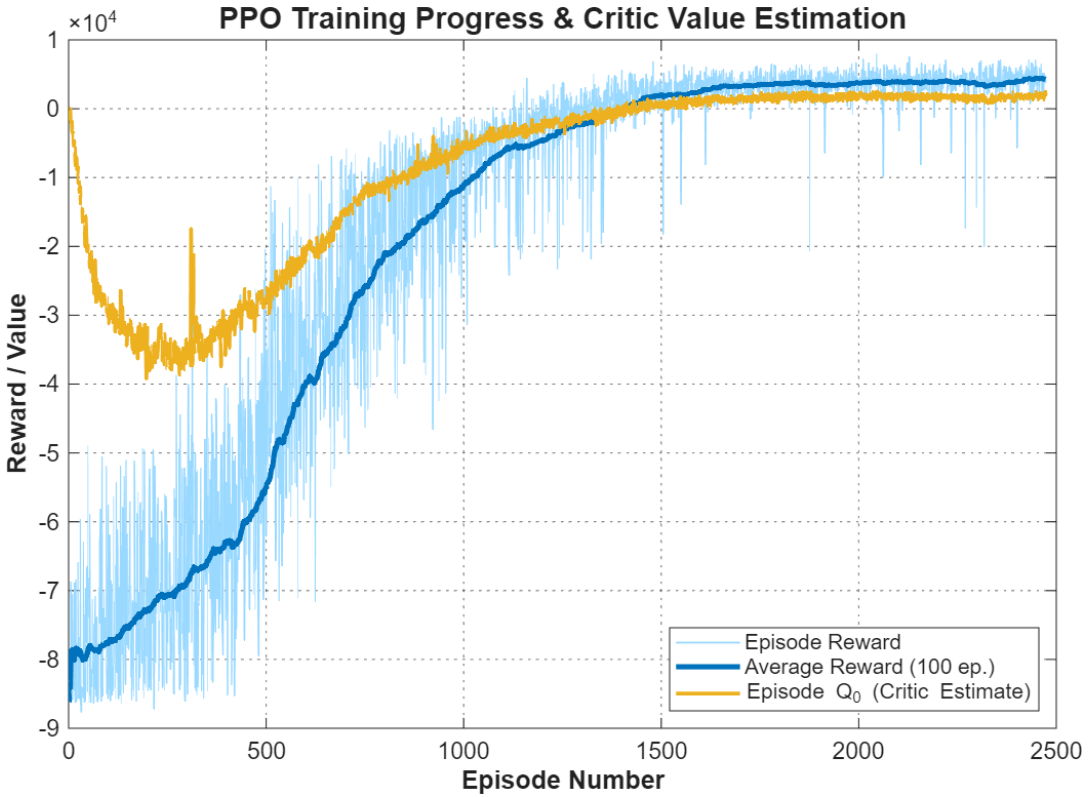


Reward function: $R(t) = R_{progress}(t) - R_{collision}(t) + R_{exit}(t)$

- $R_{\text{exit}} = 350 \cdot N_{\text{exit}}^{(t)}$: a high weighted reward proportional to the number of packages that exited.
- $R_{progress}(t) = \sum_{i=1}^{N_{active}(t)} \max(0, w_p \cdot \Delta y_i(t))$: a reward proportional to the positive displacement of each package along the y-axis at each time step with no penalty for stagnation is implemented.
- $R_{collision} = \sum_i \sum_{j>i} (\text{Overlapping}_{ij} \cdot K_p + C_{base})$: it penalizes collision proportionally to the overlapping of 2 packages and a base line penalty for the happing of a collision.

Observation state: $s_k = [x_k, y_k, d_k, o_k]^T, k = 1, \ldots, 25$ with x,y being the coordinates of the box, d the dimension of the box, o a flag of presence (all normalized).

Project 1: training (agent)

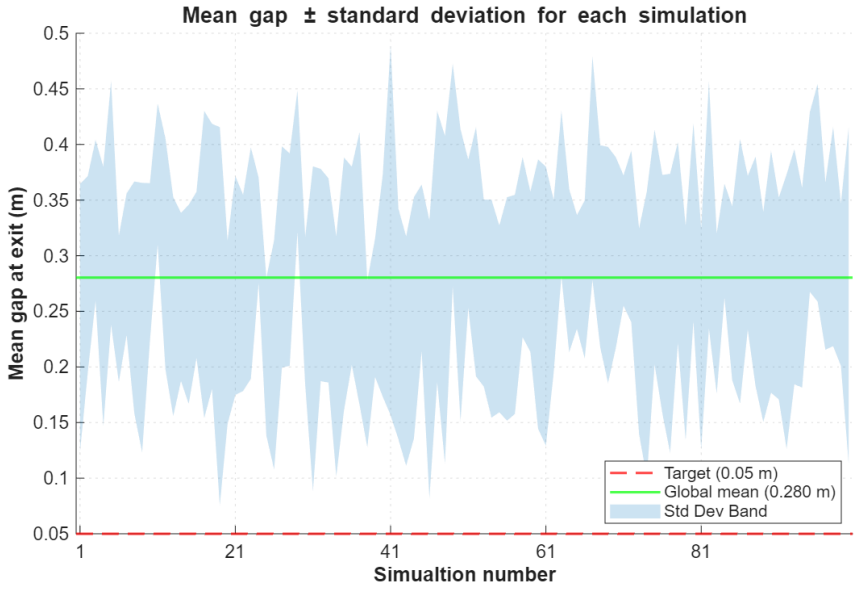


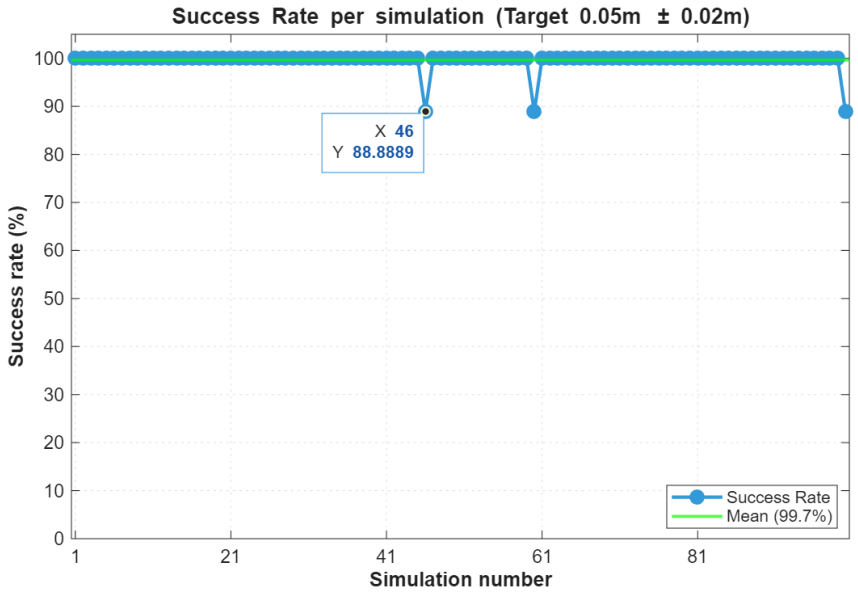
The effect of the parameters are visible on the training curves.

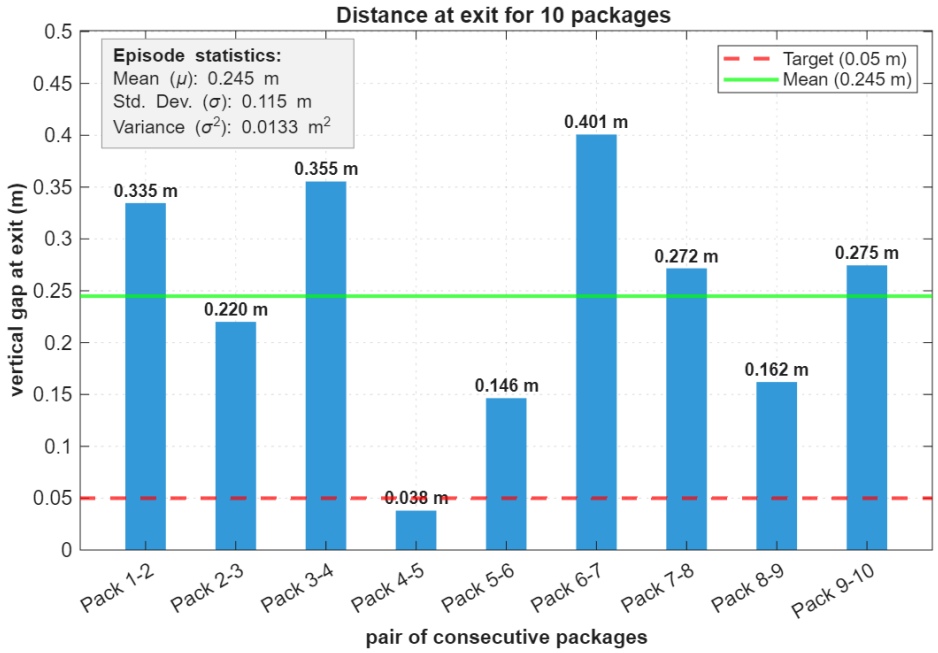
After some trials:

Experience buffer	2048	Dense sampling, gradient stability
Discount factor (γ)	0.998	Long-term perspective, final target award
GAE factor (λ)	0.95	Bias/Variance Trade-off
Learning rate actor	5e-4	Cautious Policy Update
Learning Rate Critic	9e-3	High estimate of V(s)
PPO Clipping	0.04	Limits on Policy Change, Stability
Entropy weight	0.02	Exploration Maintenance, Anti-Local Minima

Project 1: 10 packages



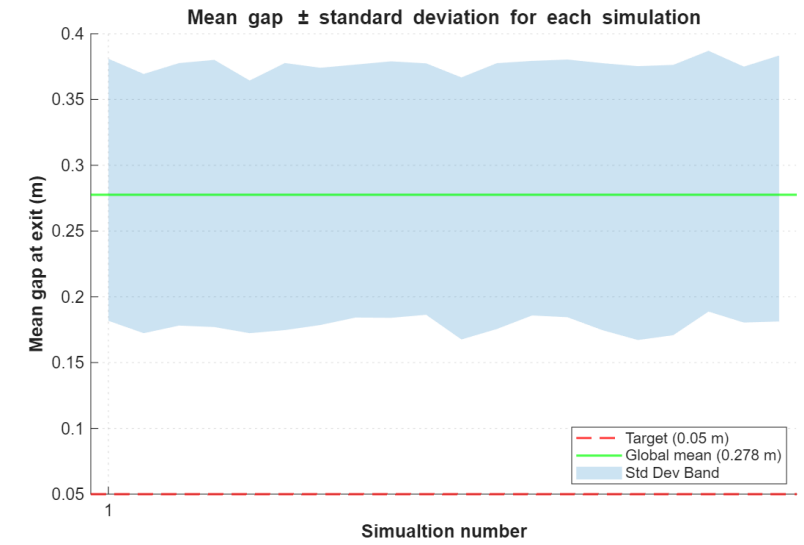


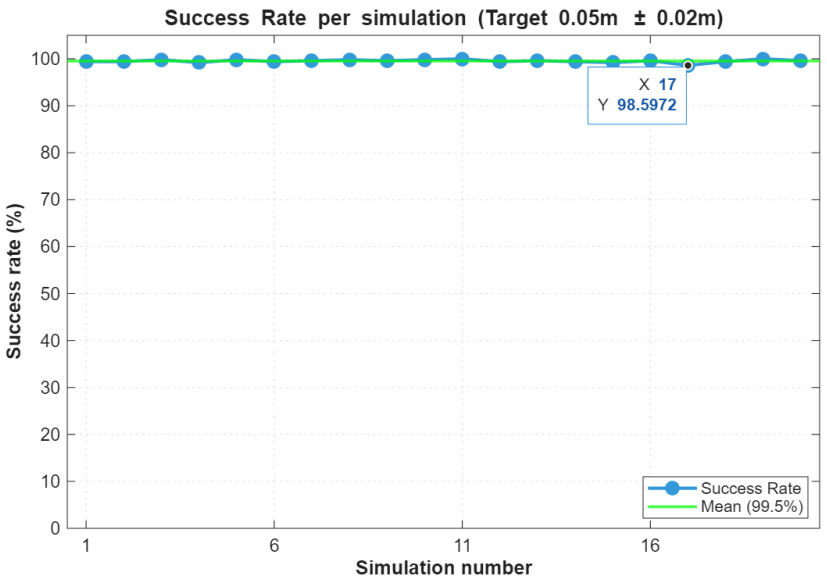


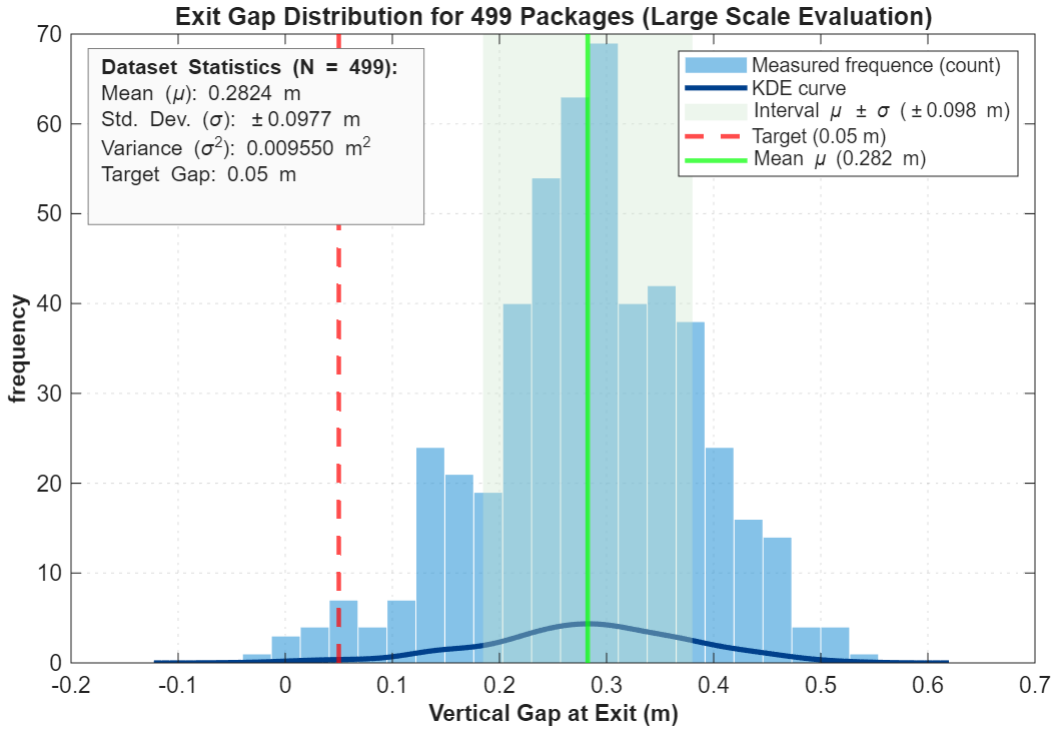
A random simulation shows an overall success rate way above the zero and the target gap.

100 simulations show the robustness of the policy with occasional drops below the mean.

Project 1: 500 packages



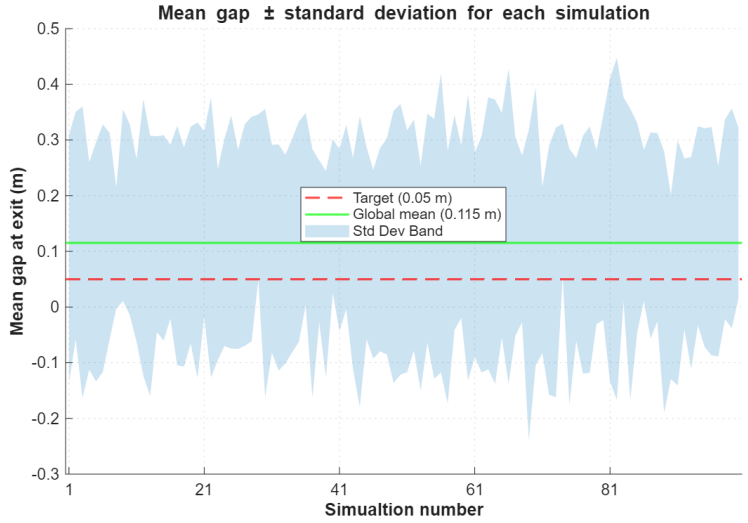


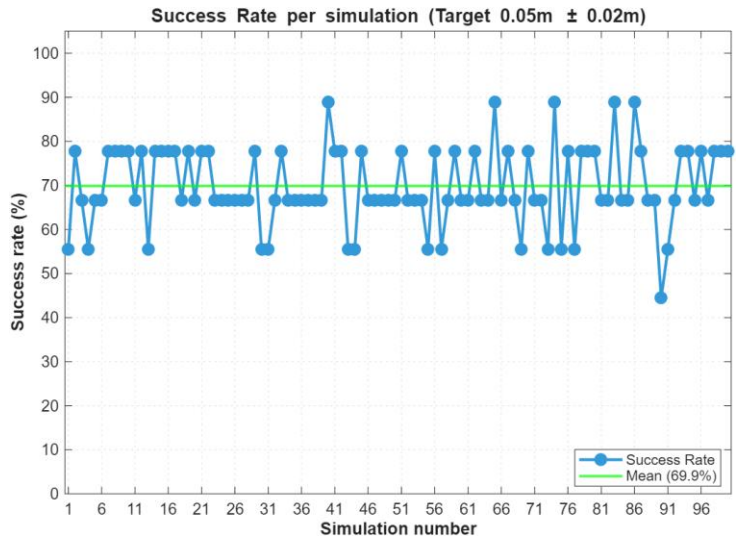


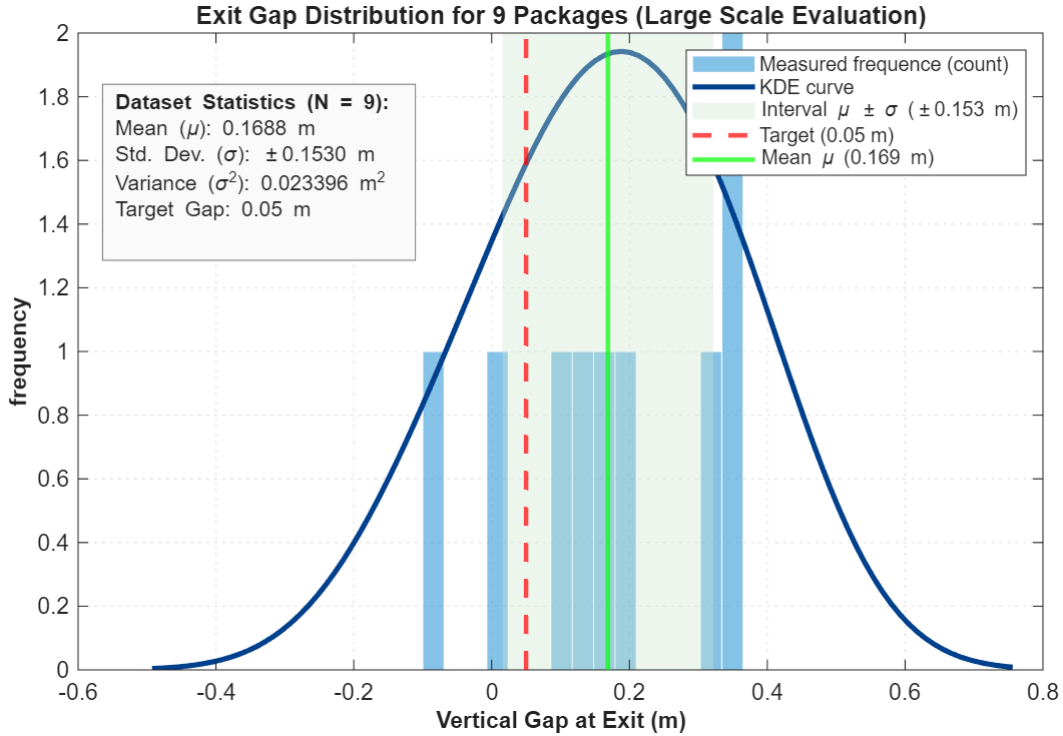
A random simulation shows the majority of boxes arrives with a mean gap of 0.3 m.

20 simulations show the robustness of the policy even with a bigger number of total packages.

Project 1: 10 packages, $t_{generation}$=0.5 s







Can the model trained under $t_{generation}$=0.75 s perform well with a faster time generation?

There is a drop in the performances across the 100 simulations. A positive mean gap is still obtained.
A retraining in this case can solve the problem and increase the perfomances!

Project 2: approaches

- **APPROACH 1: SemiBrute force method (SB)**
- **APPROACH 2: Bayesian optimization (BO)**
- **APPROACH 3: Vanilla Neural network (NN)**

2° ODE:

For the base: $M_b\ddot{x}_b + c_b\dot{x}_b + K_b x_b = K_r\left(x_r - x_{\text{cmd}}\right) + c_r\dot{x}_r$

For the robot: $M_r(\ddot{x}_b + \ddot{x}_r) + c_r\dot{x}_r + K_r x_r = K_r x_{\text{cmd}}$

And BC (position only different from zero)

x_b *is* the absolute displacement of the base

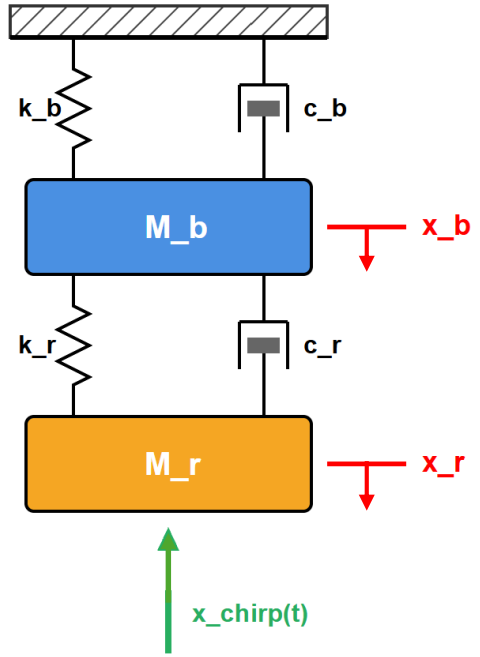
x_r is the relative displacement of the robot

Parameters: $\boldsymbol{\theta} = \left[K_b,\ K_r,\ M_b,\ h_b,\ h_r,\ f_0,\ f_1,\ A_{\text{sin}},\ x_{r,\text{start}}\right]^T$

9 parameters govern our problem

Space of search: hypercube with lower and upper bounds as:

$$\boldsymbol{\theta}_{\text{lb}}^T = [100,\ 100,\ 10,\ 0.1,\ 0.1,\ 10^{-4},\ 3,\ 0.1,\ 0.3]$$
$$\boldsymbol{\theta}_{\text{ub}}^T = [5000,\ 5000,\ 100,\ 0.3,\ 1.1,\ 10^{-2},\ 10,\ 0.2,\ 0.5]$$

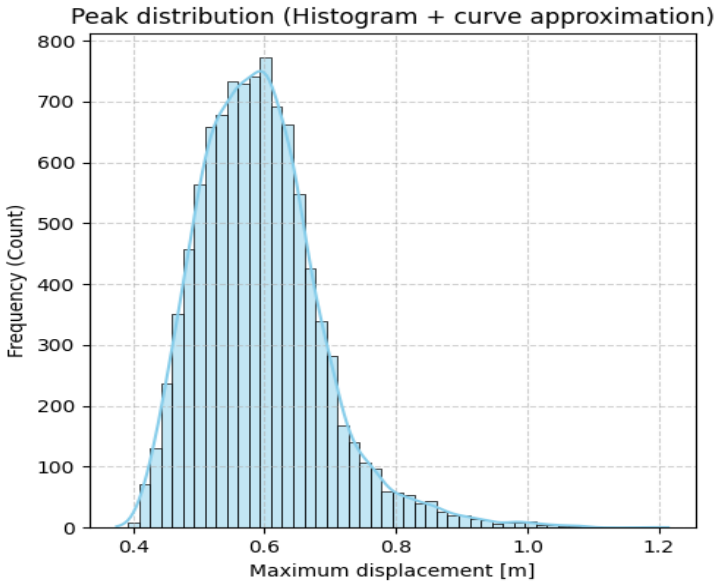


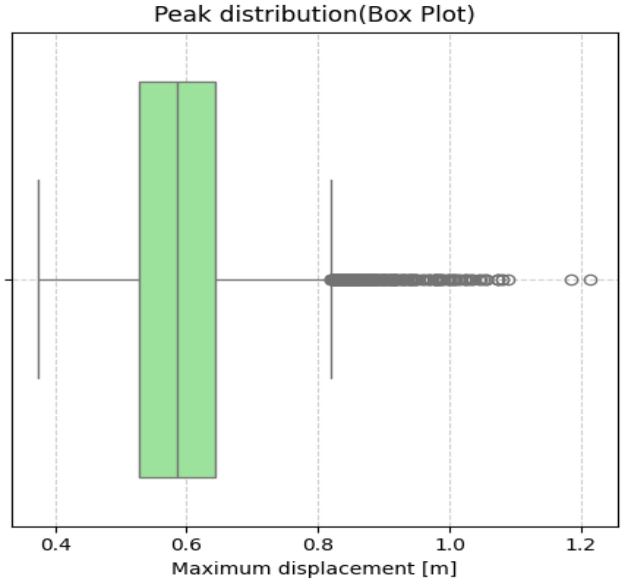
Results of SB

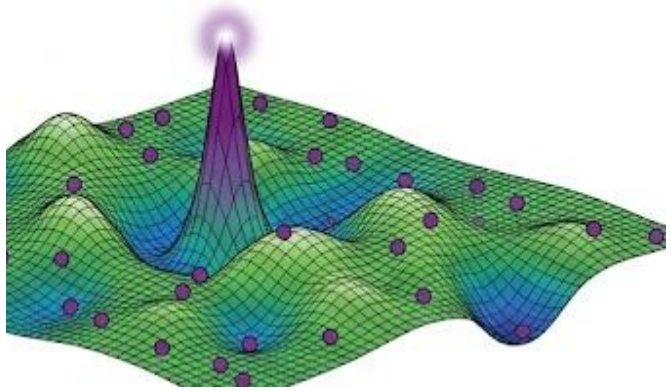
Sample the space through **LatinHypercube algorithm**:
N_samples = 10000
Tsim = 20 s
Mr = 10 kg
Tsampling = 0.05 s

Differential evolution algorithm to find the maximum: it takes time.

The inverse problem may suffer from grid resolution or, if an algorithm is used, it also takes time.







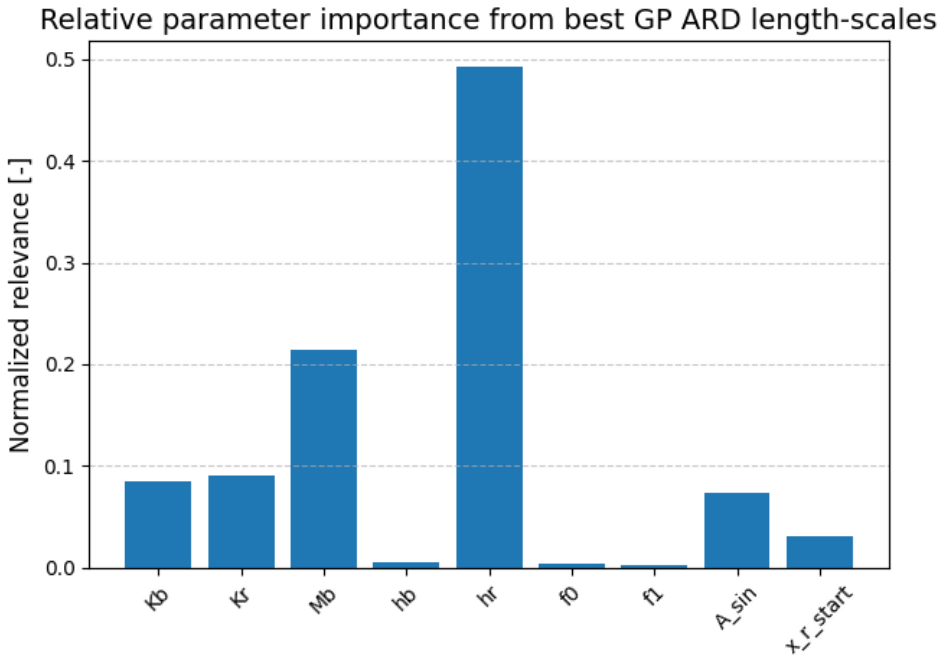
Results of SB: direct and inverse problem

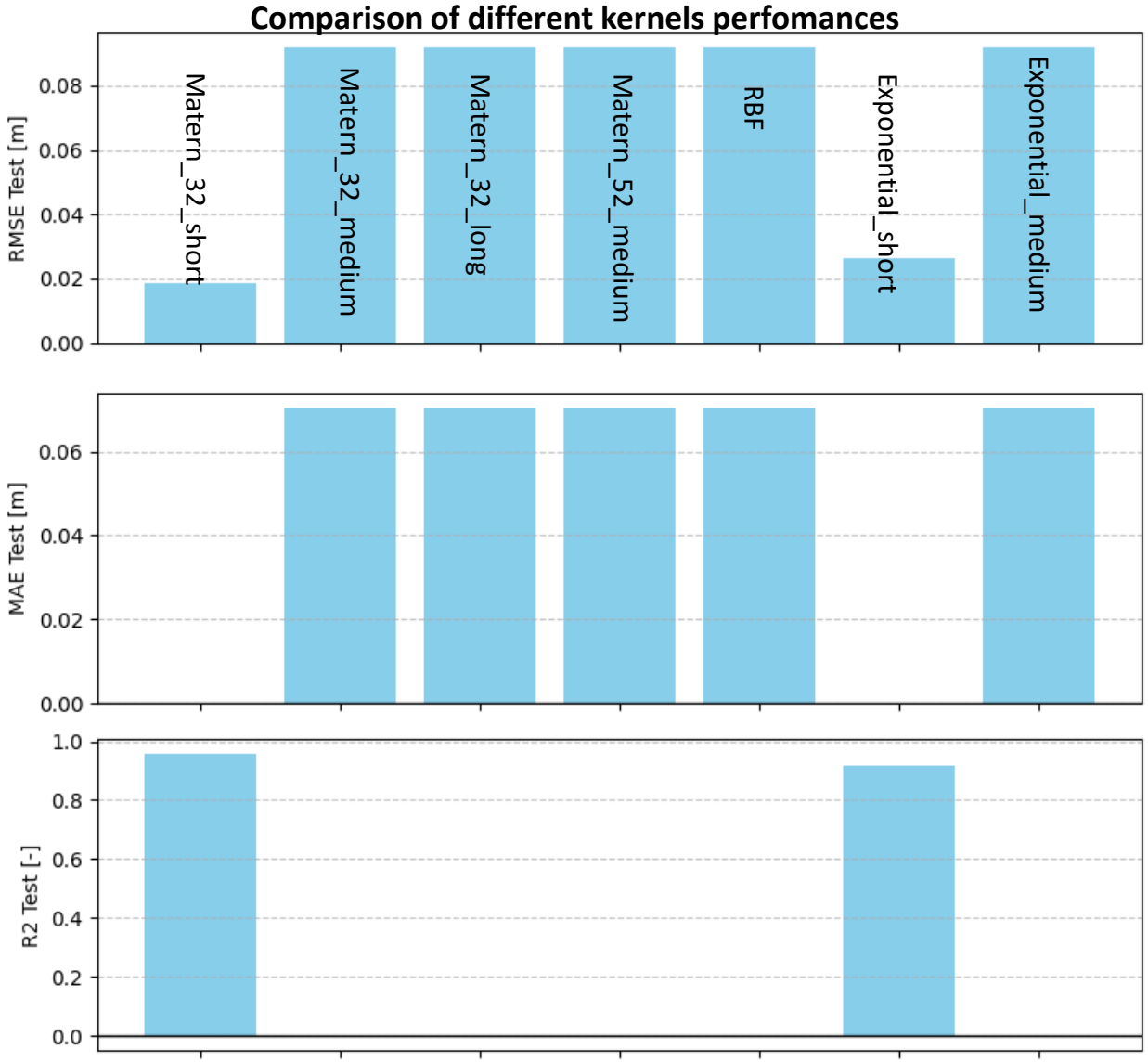
Parameter	Optimal Value
K_b[N/m]	4889.12
K_r[N/m]	2617.95
M_b[kg]	10.0
h_b[-]	0.1
h_r[-]	0.1
f_0[Hz]	0.0001
f_1[Hz]	3.0
$A_{\sin}$[m]	0.2
$x_{r,\text{start}}$[m]	0.5
Maximum displacement of the robot [m]	1.577218

Parameter / Metric	Value
K_b[N/m] *(FIXED)*	4889.12
K_r[N/m] *(OPTIMIZED)*	4315.18
M_b[kg] *(FIXED)*	10.00
h_b[-] *(FIXED)*	0.10
h_r[-] *(OPTIMIZED)*	0.12
f_0[Hz] *(OPTIMIZED)*	0.0029
f_1[Hz] *(OPTIMIZED)*	9.35
$A_{\sin}$[m] *(OPTIMIZED)*	0.17
$x_{r,\text{start}}$[m] *(OPTIMIZED)*	0.49
Target displacement [m]	1.12
Achieved peak displacement [m]	1.12002

- The solution is hitting the boundaries of our space.
- Time consuming: the search for max requires almost 5 minutes.
- The inverse problem requires 1.4 minutes for 1 simulation!

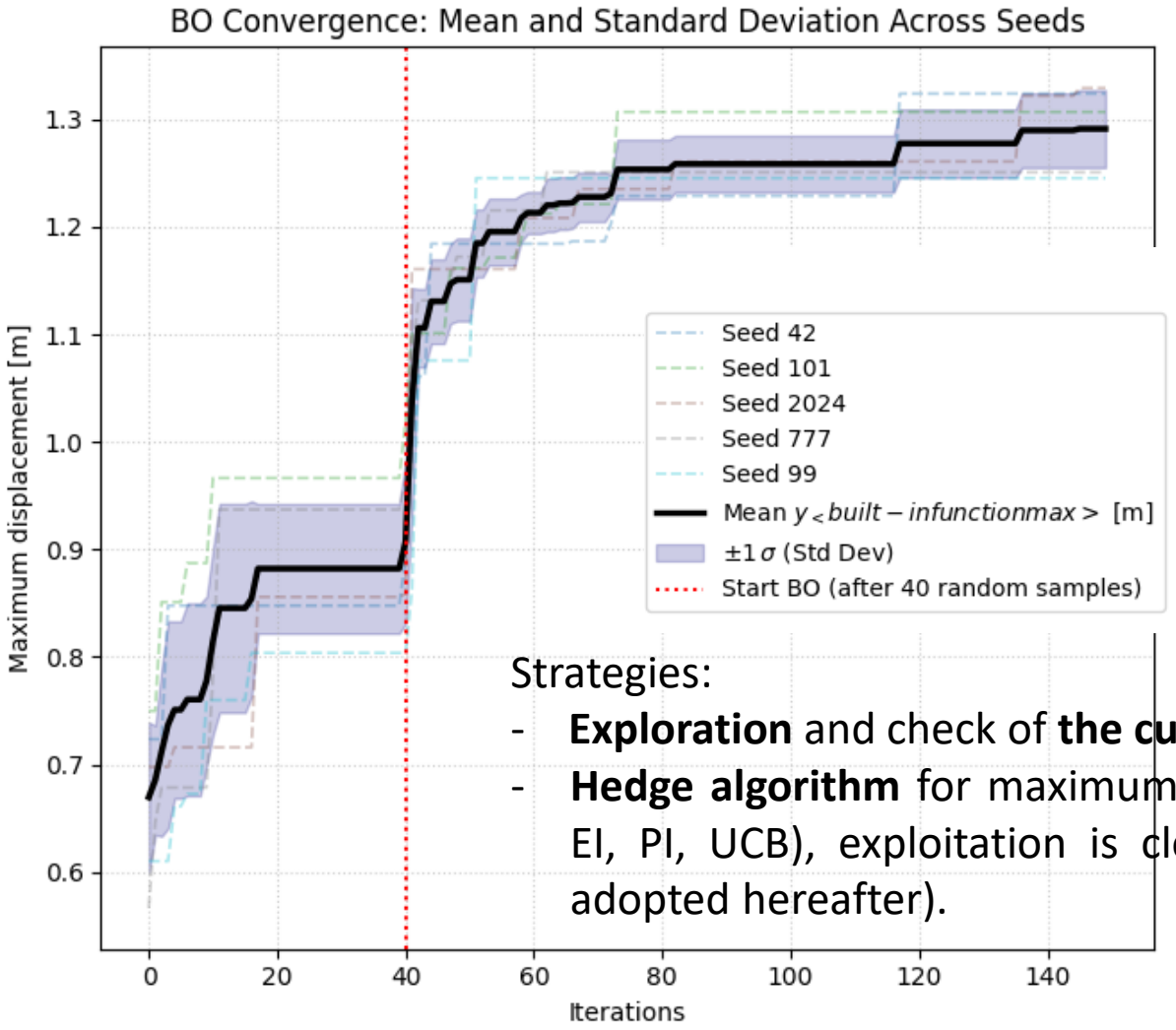
Results of BO: benchmark

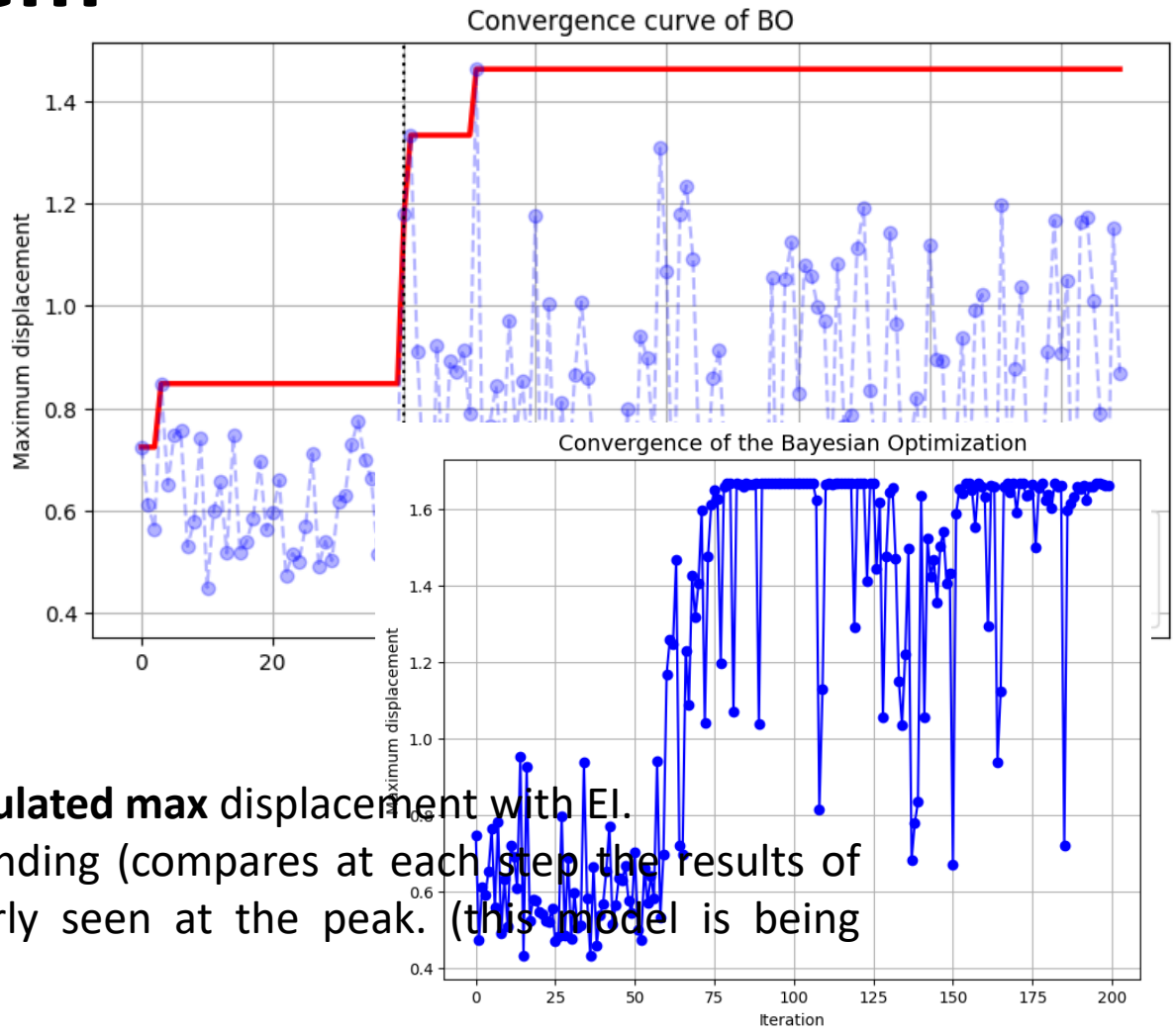




- A Matern 32_ short was selected thanks to a benchmark analysis.
- The importance of the lenghtscale is also shown: notice the obvious importance of the adimensional damping of the robot.

Results of BO: direct problem

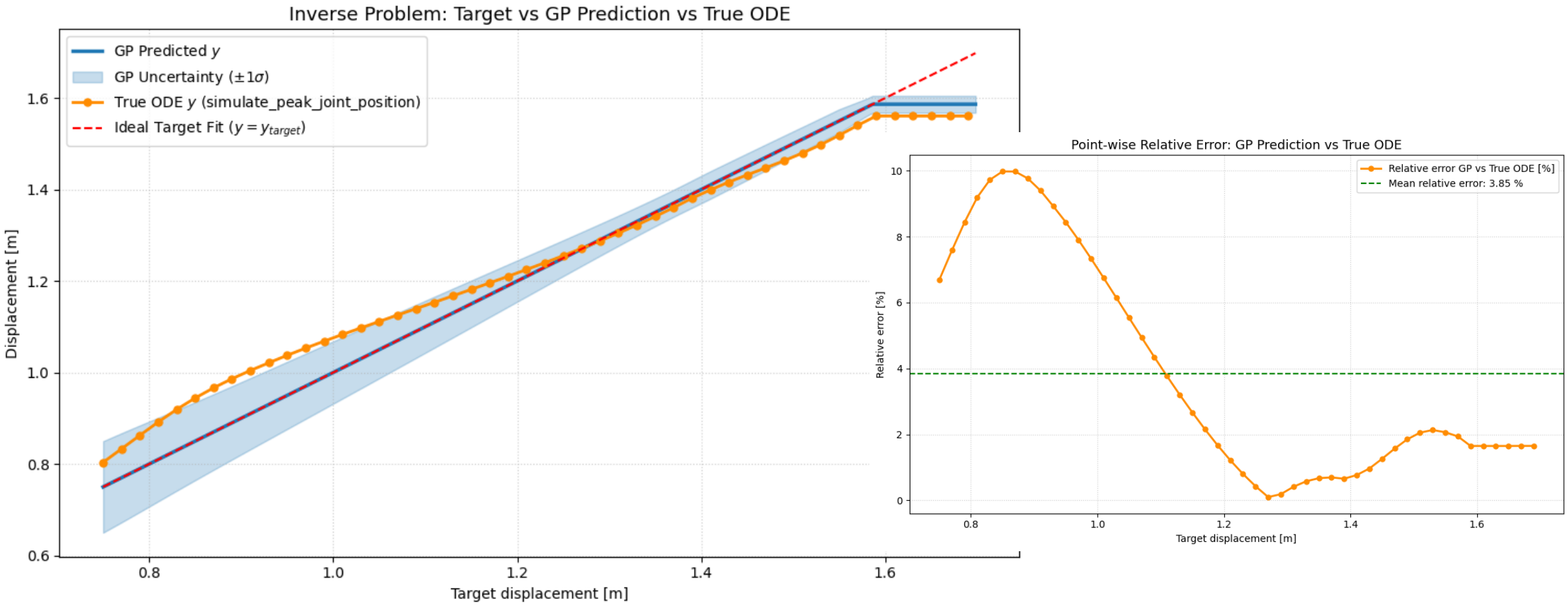




Strategies:

- **Exploration** and check of **the cumulated max** displacement with EI.
- **Hedge algorithm** for maximum finding (compares at each step the results of EI, PI, UCB), exploitation is clearly seen at the peak. (this model is being adopted hereafter).

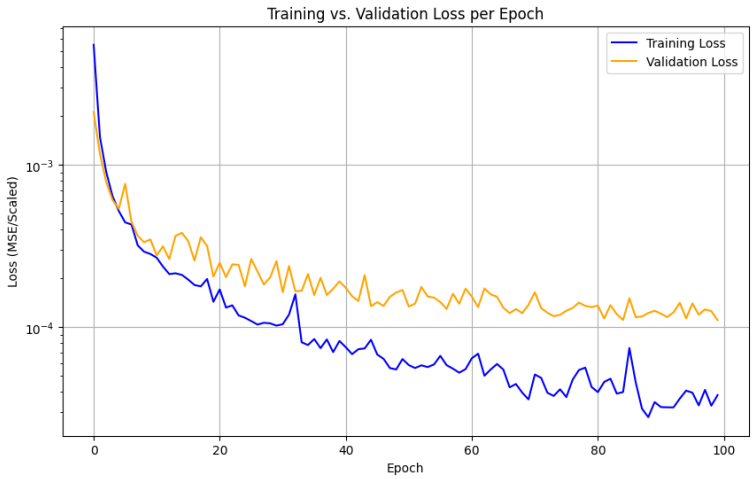
Results of BO: inverse problem



The uncertainity is lowering as y_{target} increases exploitation takes place.

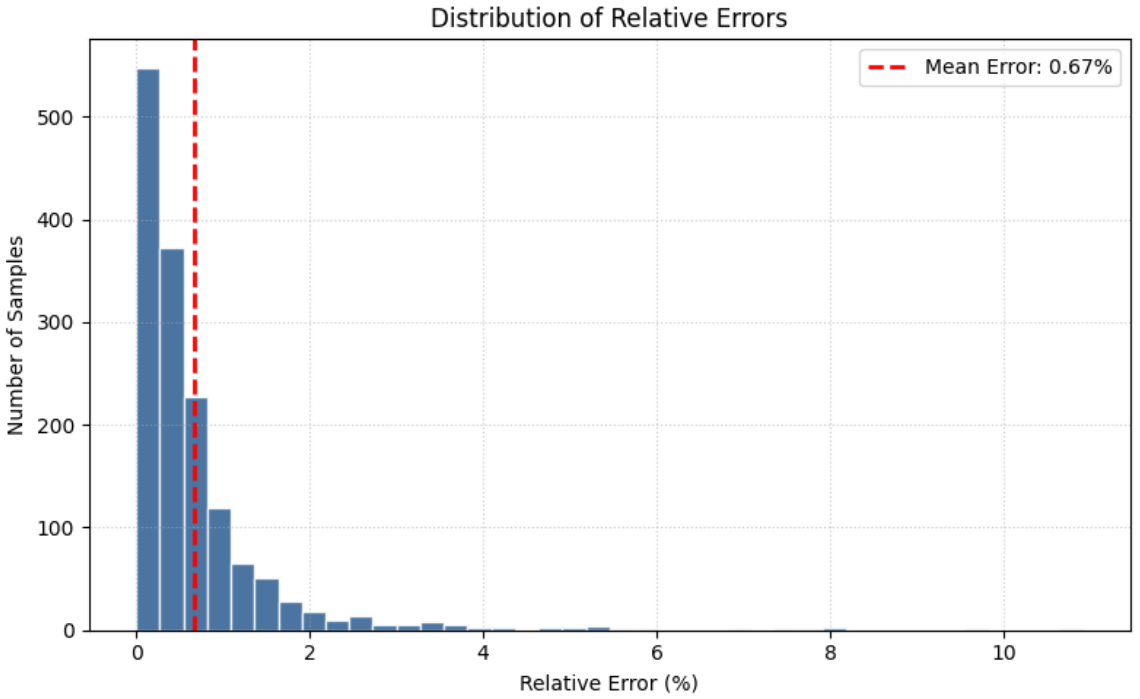
The plateau happens either for insufficient base parameters or unexplored region.

Results of NN: training the net



The loss is being minimized. No overfitting is shown. The error on the test set is low.

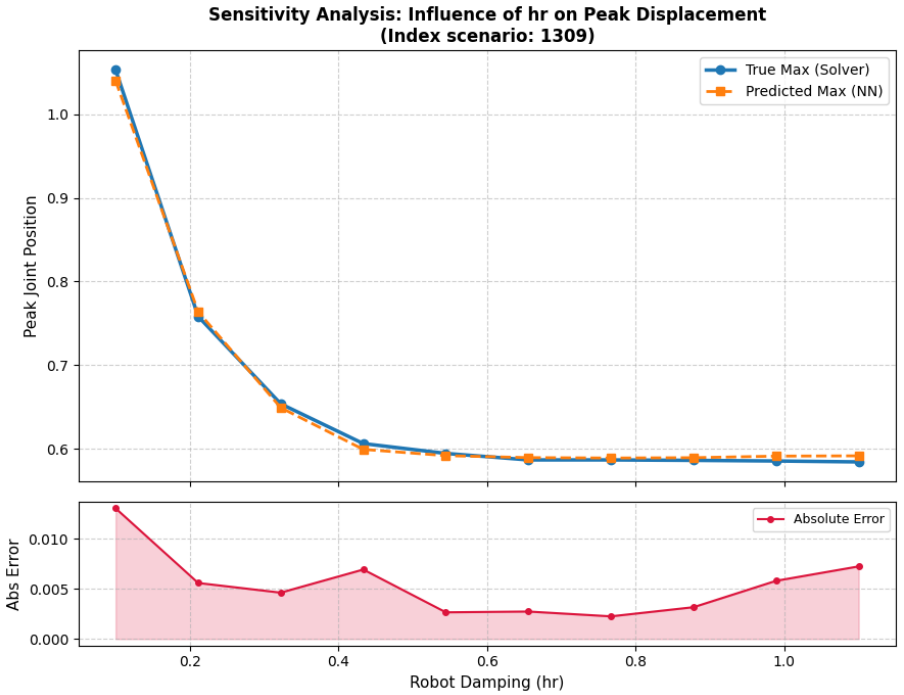
The architecture being used is the one on the right





Results of NN: direct problem

Parameter	Value
K_b[N/m]	1938.5537
K_r[N/m]	2303.4365
M_b[kg]	94.3593
h_b[-]	0.1166
h_r[-]	0.1010
f_0[Hz]	0.0048
f_1[Hz]	5.4579
$A_{\sin}$[m]	0.1989
$x_{r,\text{start}}$[m]	0.4928
Maximum displacement of the robot [m]	1.179597



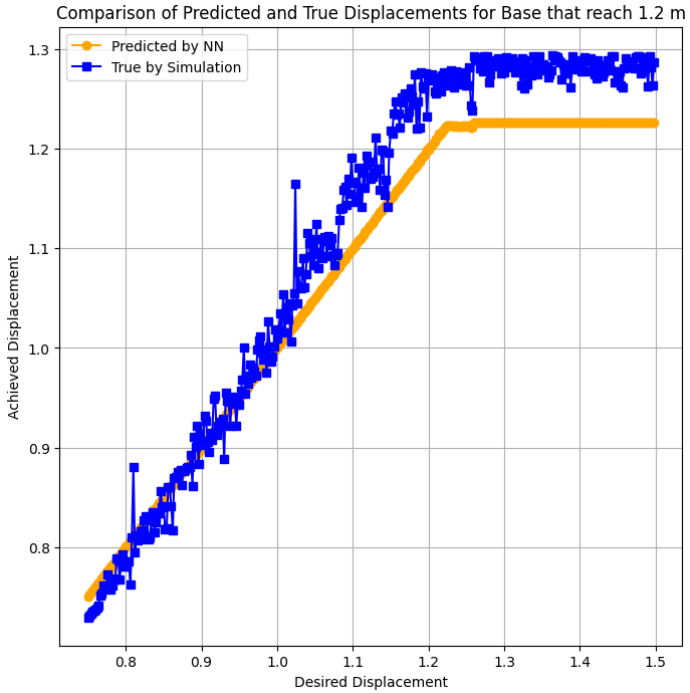
The NN has really learnt as the sensitivity analysis on h_R shows.

Levaraging the surrogate model leads to ultra fast compuatation of the order of milliseconds.

For maximum search a simple **Monte carlo approach** is used with 500_000 points. (still few seconds of running!)

Clearly other minimization approaches are possible (e.g. differential evolution or Autograd+Adam)

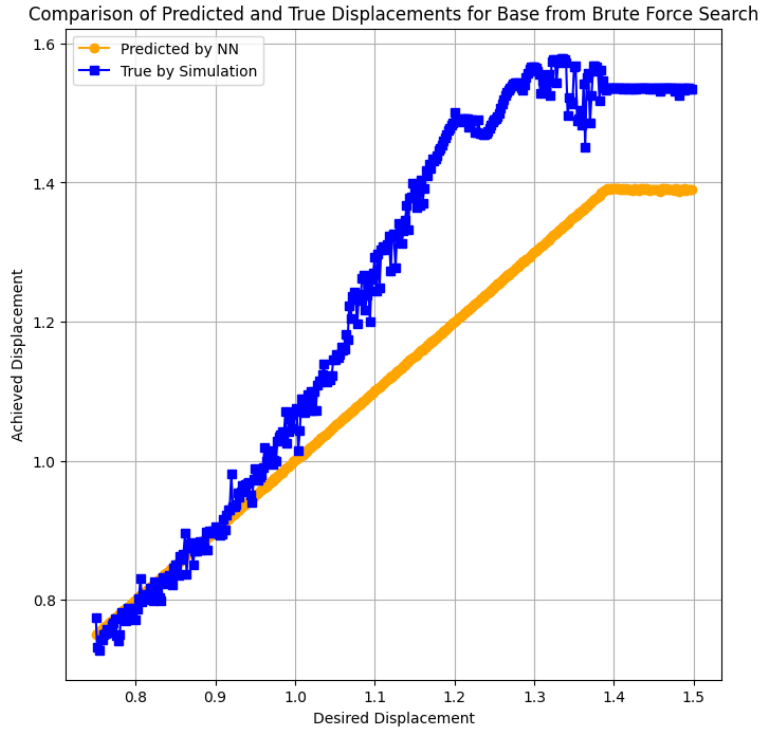
Results of NN: inverse problem

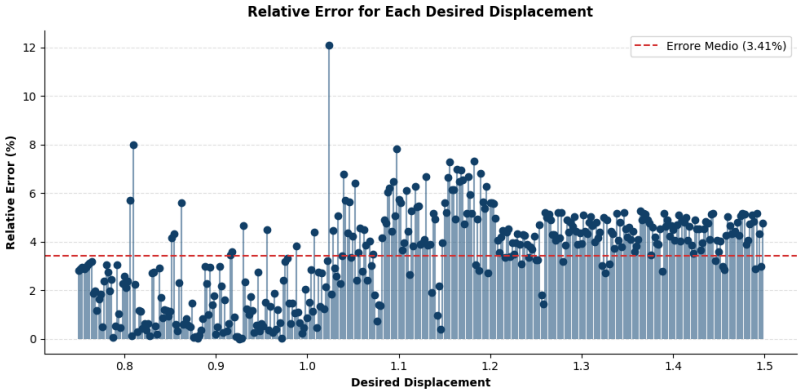


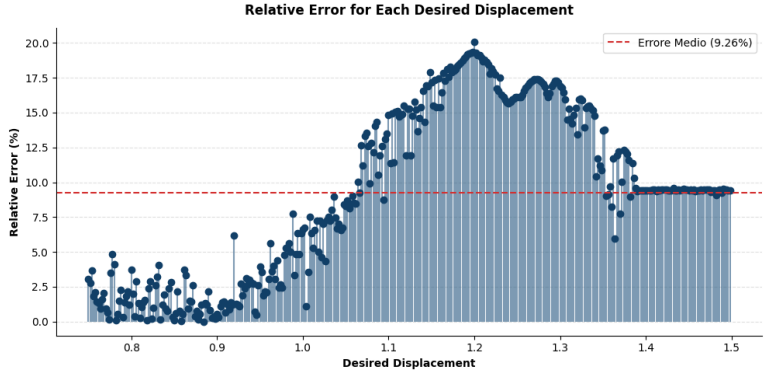
Accuracy is high where data fall in the training data distribution.

To increase accuracy do a refined sampling where the peak is.

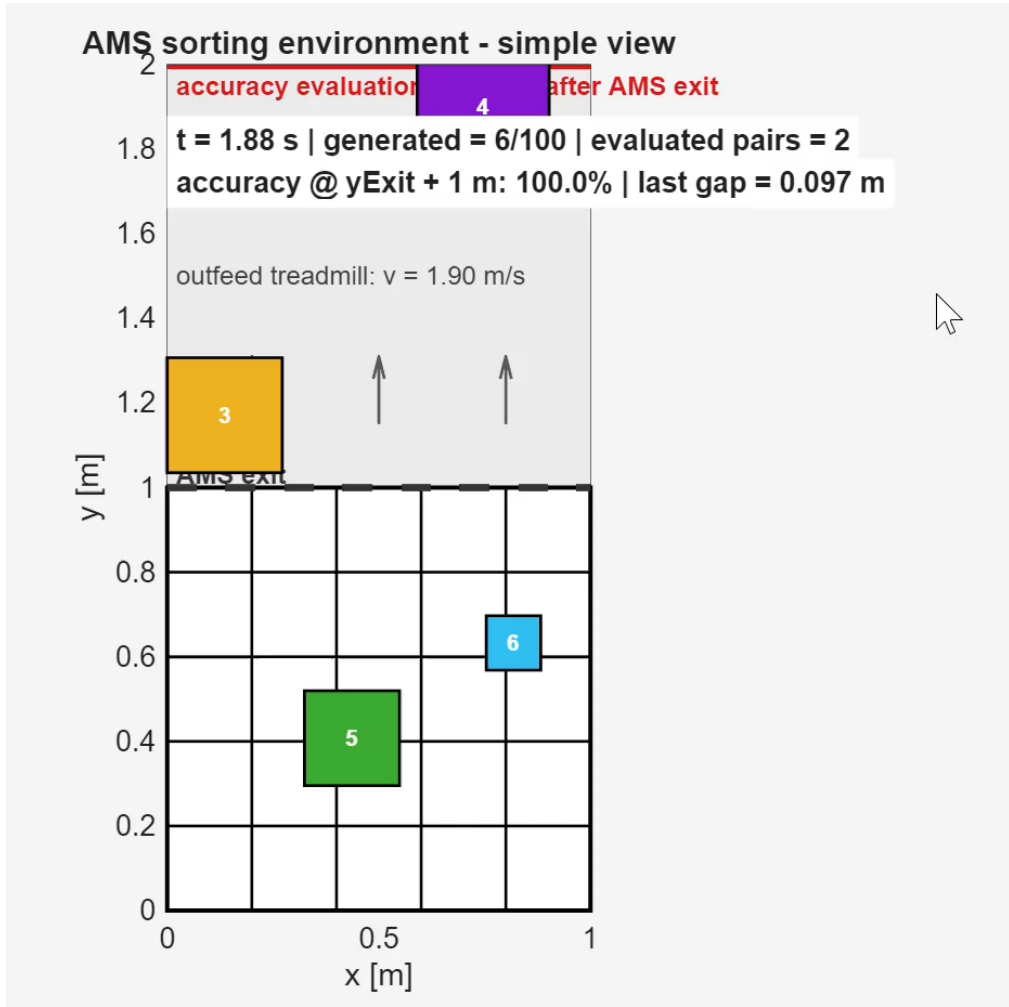
The plateau is either due to the base selected being not suitable or insufficient data being adopted.







END


AMS sorting environment - simple view
accuracy evaluation
after AMS exit
4
t = 1.88 s | generated = 6/100 | evaluated pairs = 2
accuracy @ yExit + 1 m: 100.0% | last gap = 0.097 m
outfeed treadmill: v = 1.90 m/s
3
AMS exit
6
5
y [m]
x [m]
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
0.5